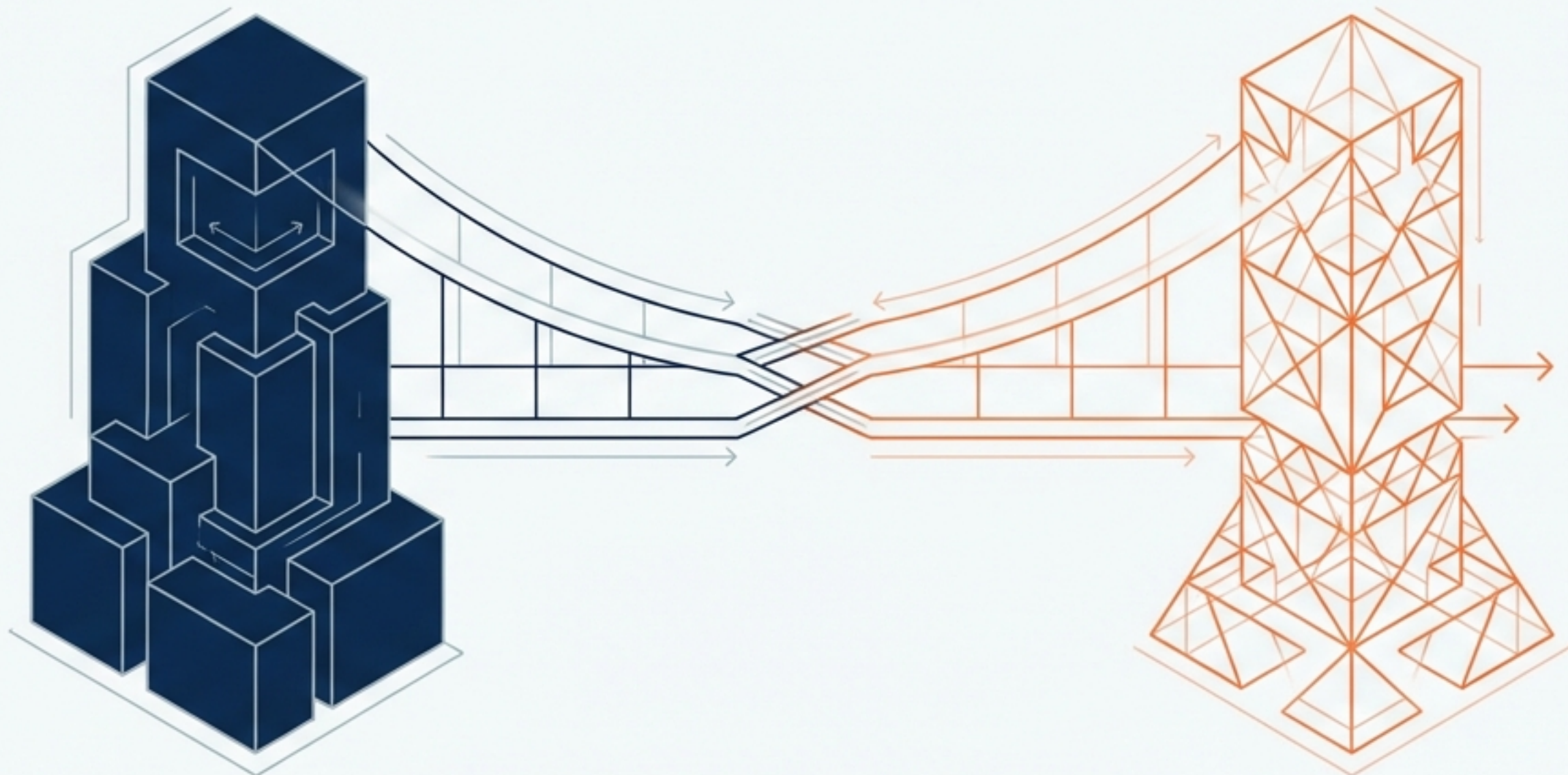


Executive Strategic Vision

Bridging Open-Source Agility and Ed-Fi Standards

Breaking the False Dichotomy—Standardization without Stagnation.



Open Education Analytics | OSS Framework

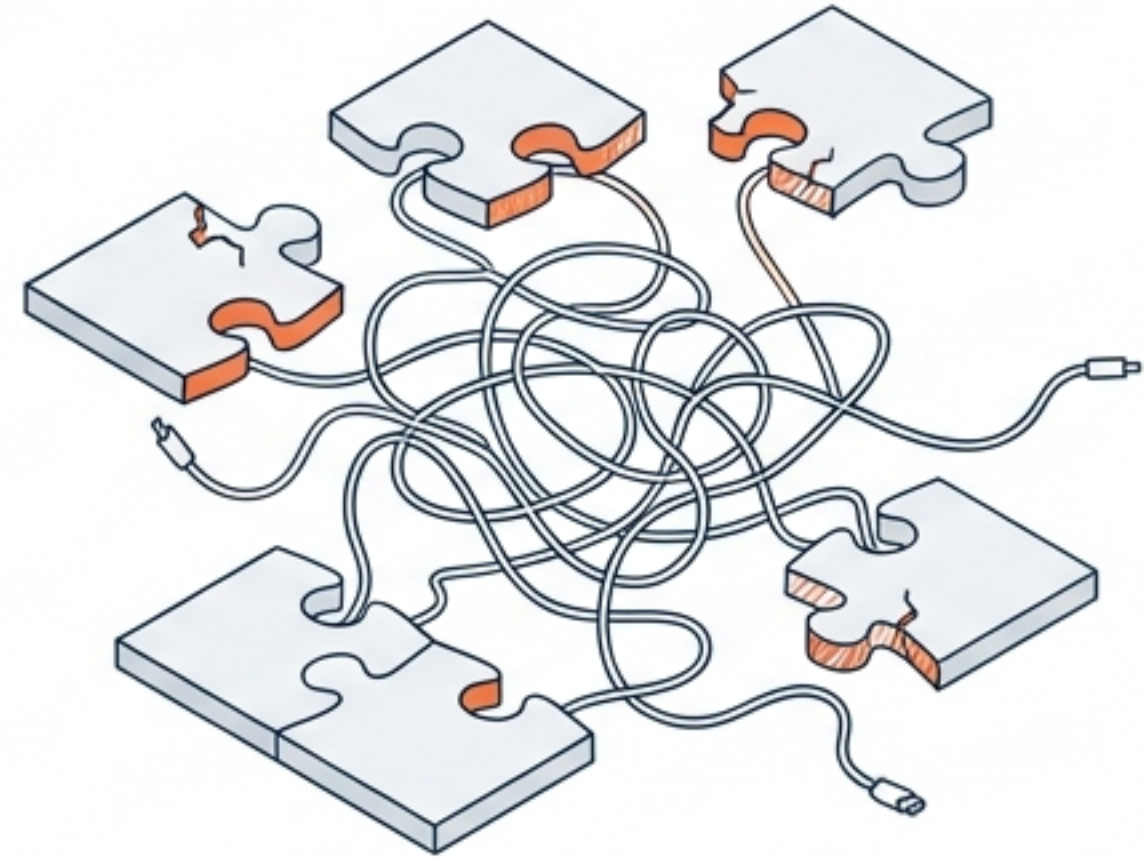
The Current Trap: A Choice Between Bankruptcy and Bad Data

The Enterprise Trap



- High Cost Infrastructure (\$188k+/yr)
- Heavy Cloud Dependencies (Azure/AWS)
- Slow Implementation (12-18 Months)

The Silo Trap



- Fragmented Reality
- Proprietary Vendor Lock-in
- “Dark Data” (Collected but unused)

The False Dichotomy: Districts are forced to choose between **unsustainable costs** or **unusable silos**.

The Vision: Democratized Interoperability



Global Standards (Ed-Fi)



Local Agility (OSS)



Sustainable Intelligence

Shift 1:

From “Heavy Iron” Servers
→ To Lightweight Logic

Shift 2:

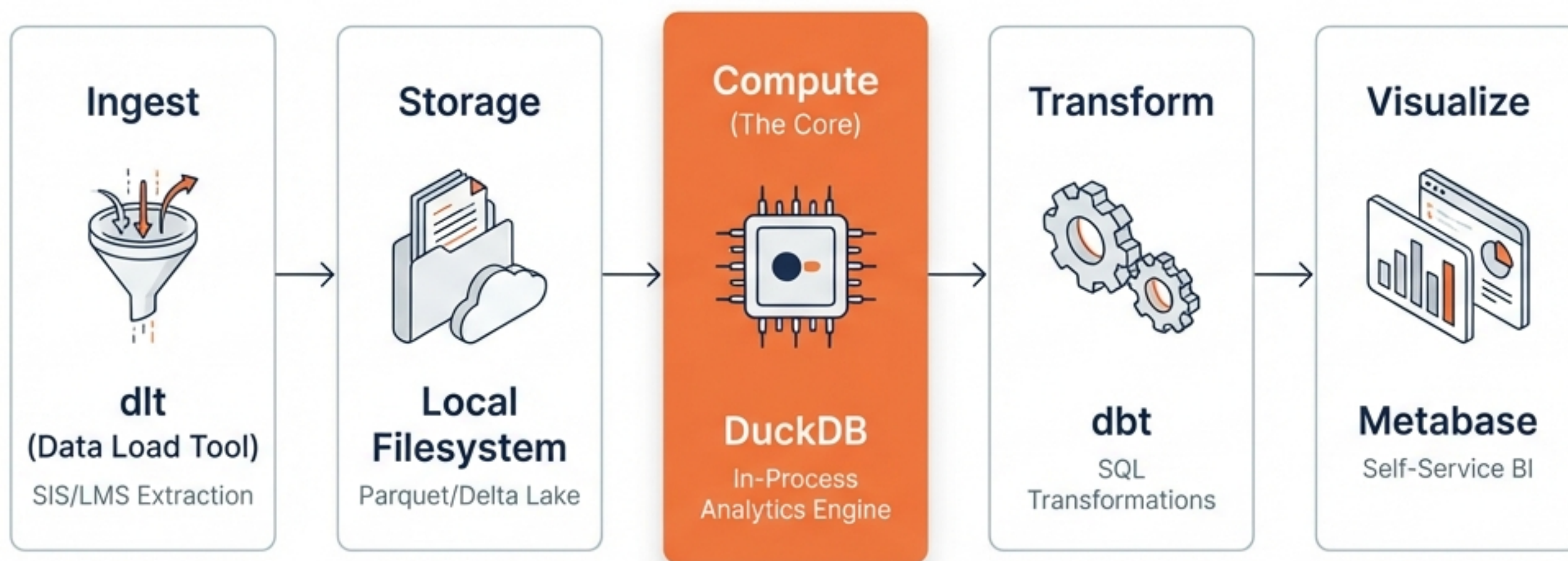
From Vendor Lock-in →
→ To **Open Data Estate**

Shift 3:

From Reporting Lag →
To Real-time Insight

The "DuckLake" Architecture

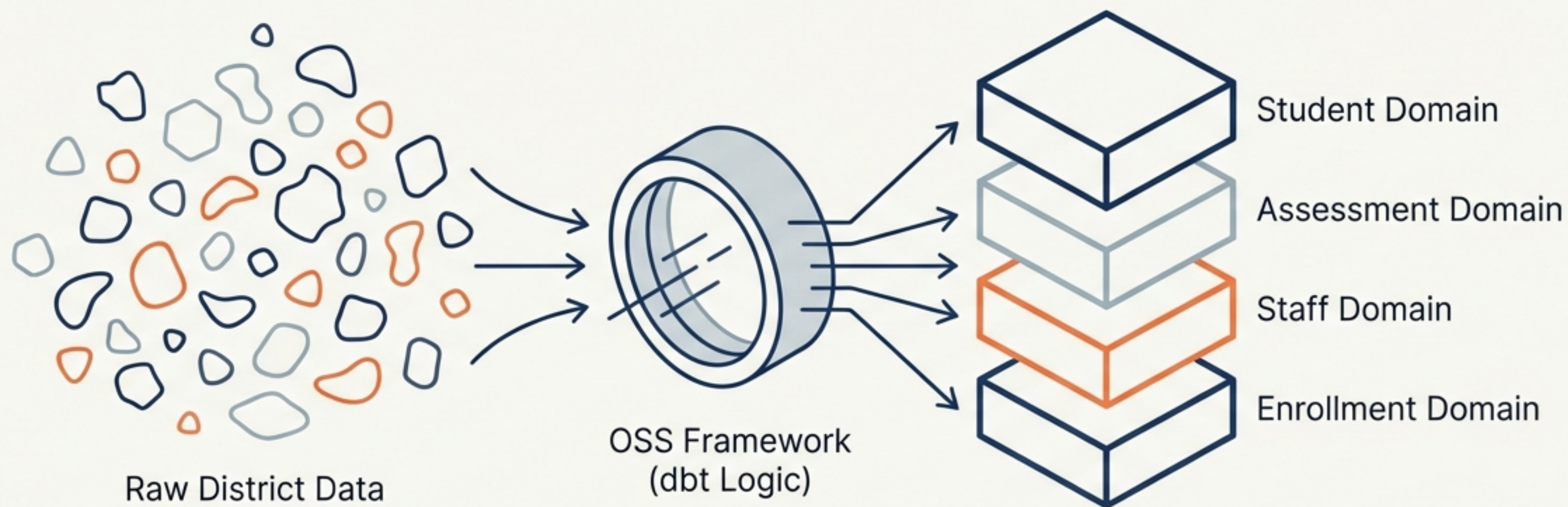
Stripping away cost, not capability.



Performance at Scale:

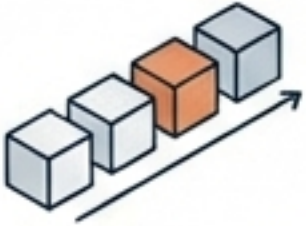
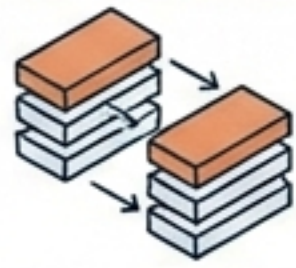
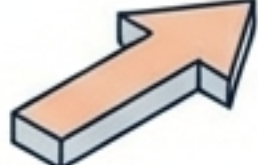
- **10x Faster** queries via Vectorized Execution
- **Zero Licensing** costs (100% Open Source)
- **Single Server** efficiency (Commodity Hardware)

The Secret Sauce: Logical Compliance, Physical Agility



We utilize Ed-Fi's definitions (Unifying Data Model) without incurring the overhead of their traditional ODS infrastructure. Your data speaks the national language, but lives in your local, agile estate.

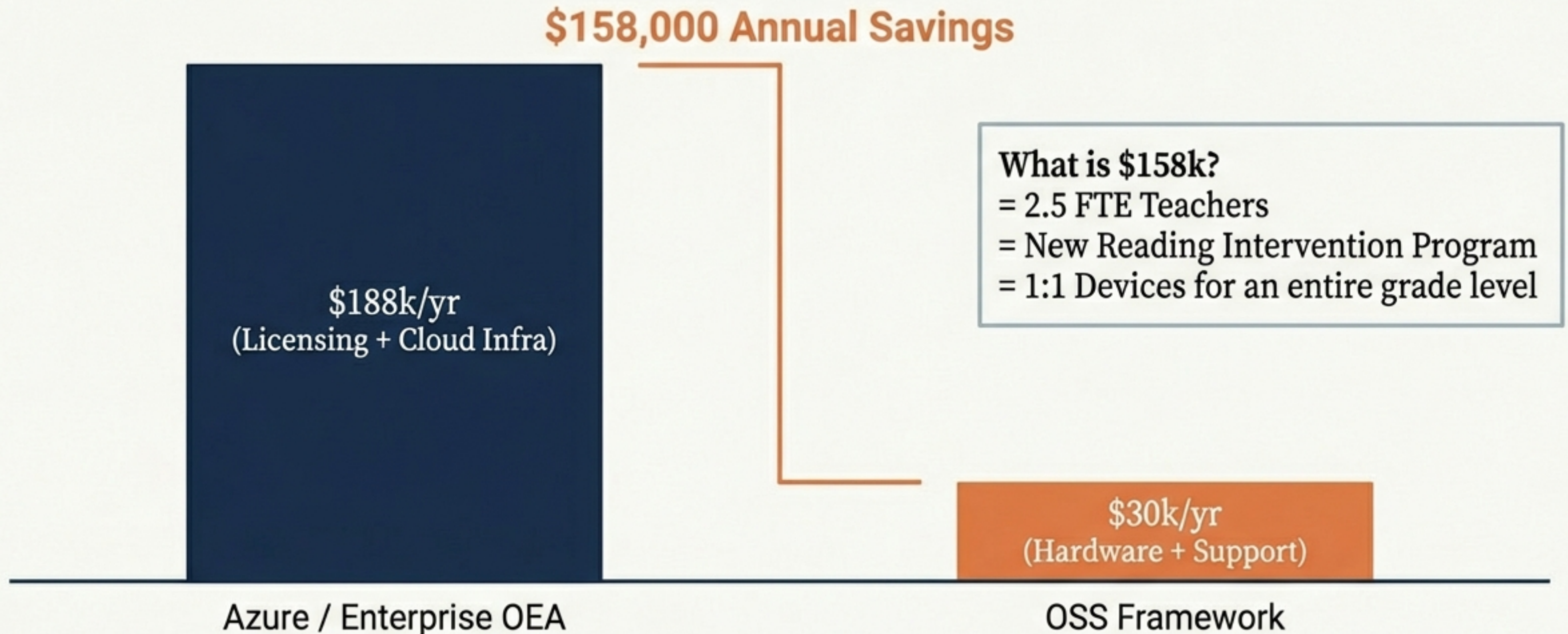
Bypassing the Speedbumps: ODS vs. Analytics Estate

Traditional Ed-Fi ODS	OSS Framework (DuckDB)
 Workload: Transactional (OLTP)	 Workload: Analytical (OLAP)
 Engine: Row-based (Slow for aggregations)	 Engine: Columnar & Vectorized (Blazing fast reads)
 Infrastructure: Heavy (SQL Server / API Clusters)	 Infrastructure: Serverless / Embedded
 Role: The Speedbump to Insight	 Role: The Accelerator for Insight

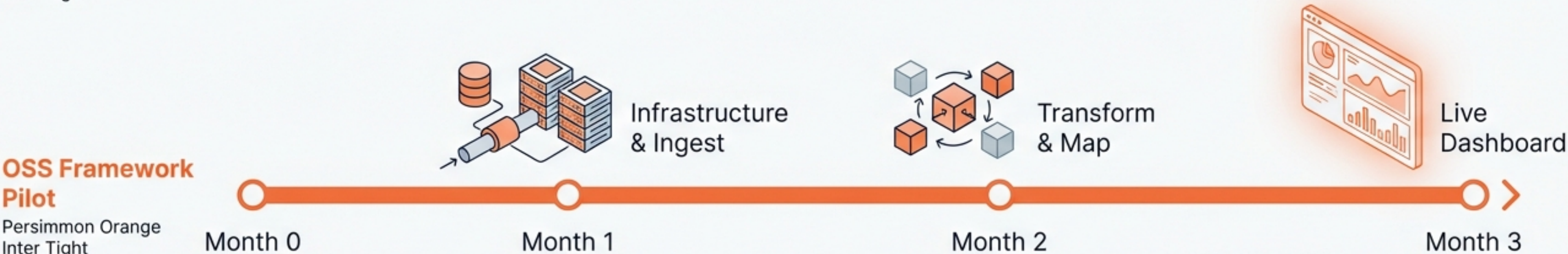
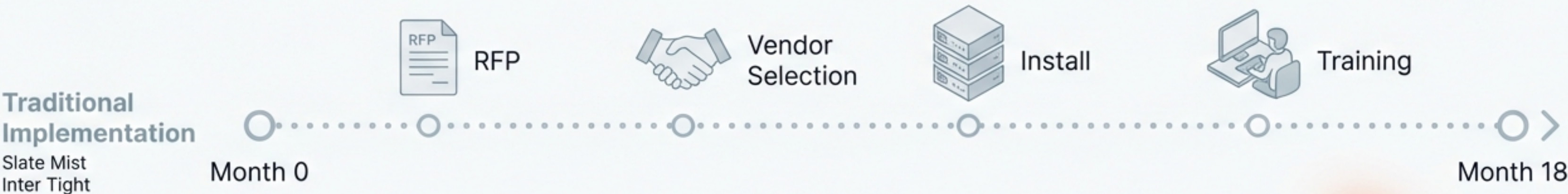
The Shift: Moving from rigid 'Schema-on-Write' to flexible 'Schema-on-Read'.

The Financial Revolution

Reallocating budget from software licenses to student learning.



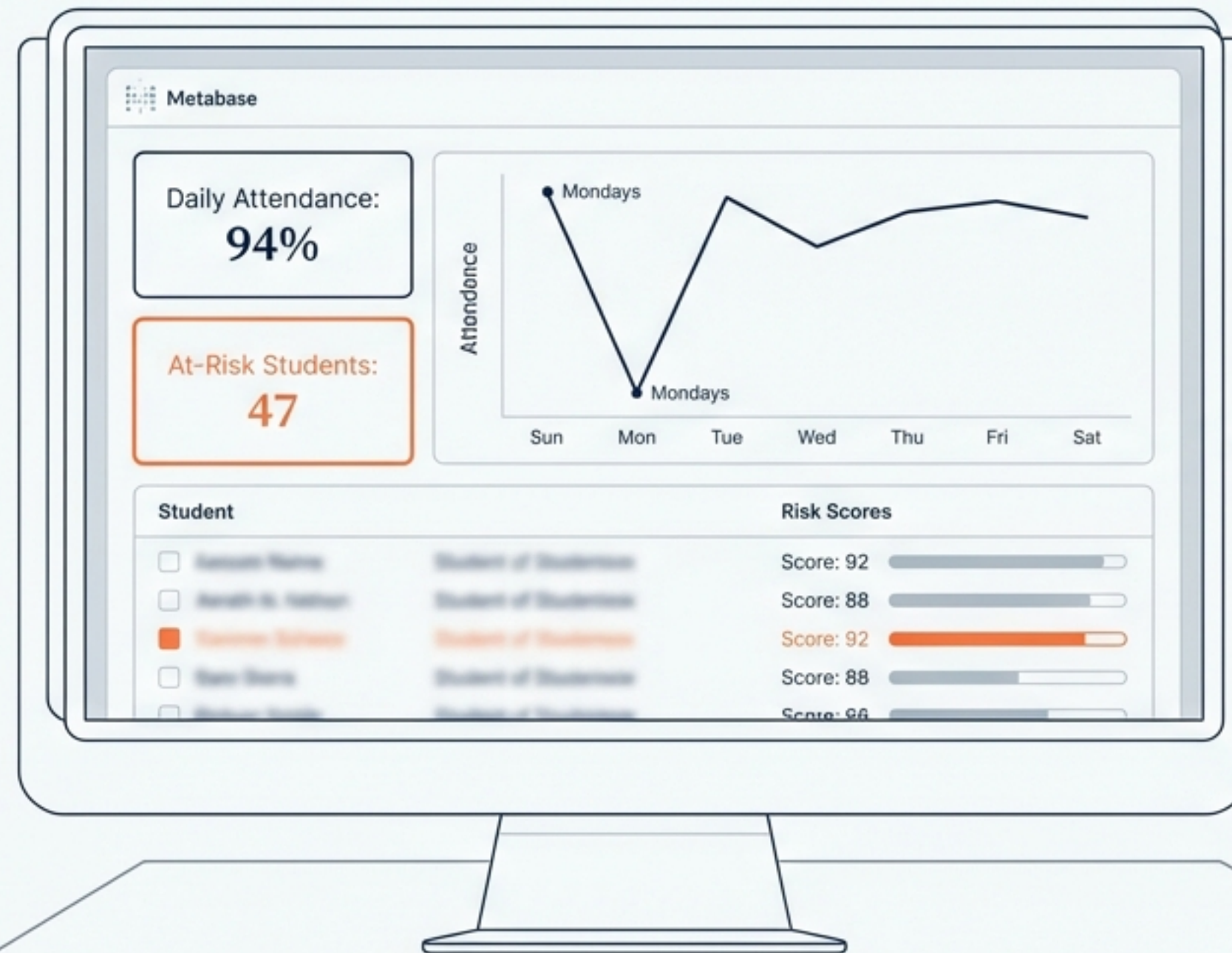
Speed to Insight: From Years to Weeks



Don't wait for the perfect warehouse.
Start solving problems now.

Use Case: Predictive Intervention

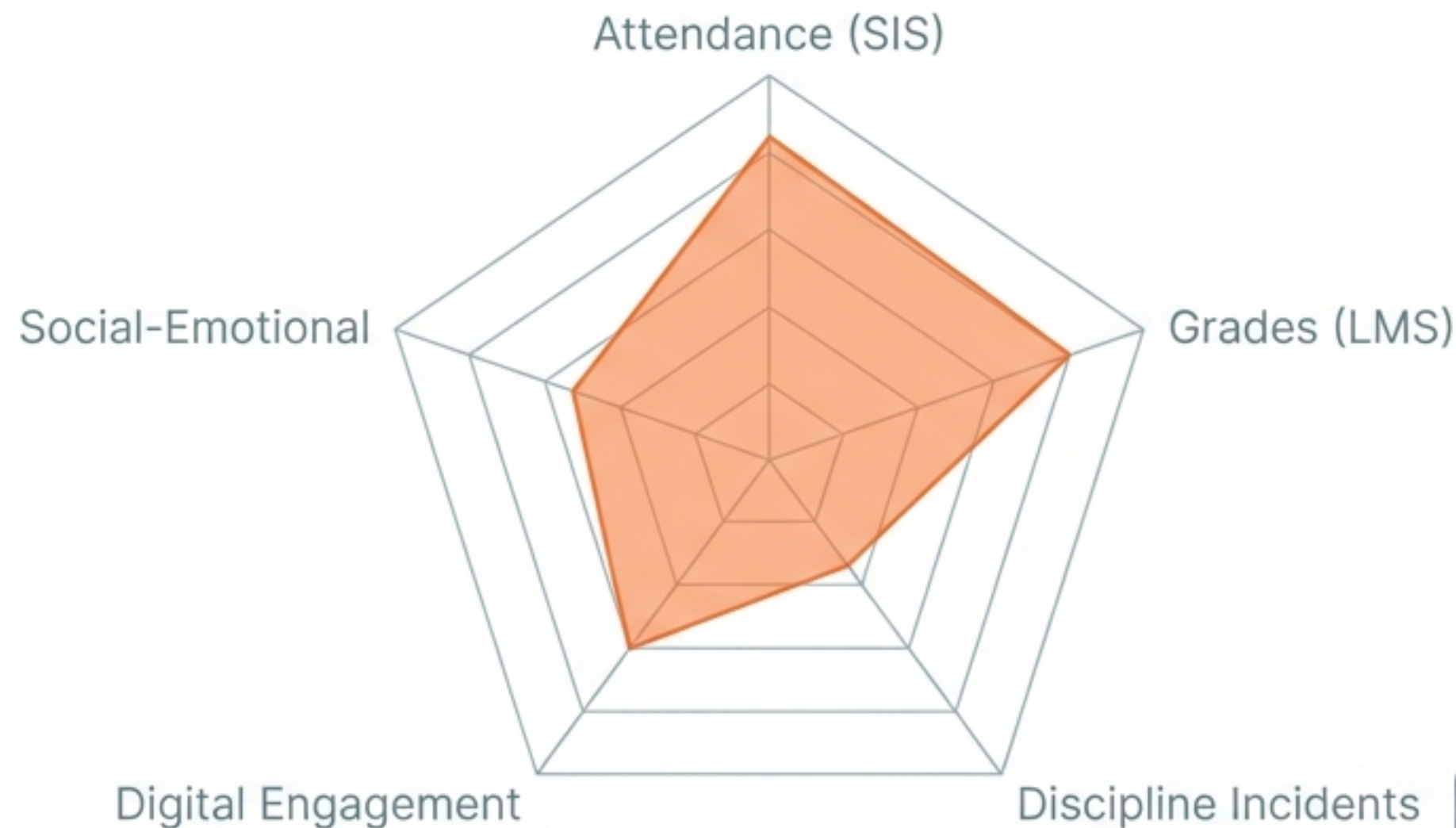
In Sub-headline: Solving Chronic Absenteeism



Real-World Pilot Results:

- Graduation Rate improved 78% → **85%**
- Chronic Absenteeism **reduced 15%** in Q2

Use Case: Equity & The Whole Child

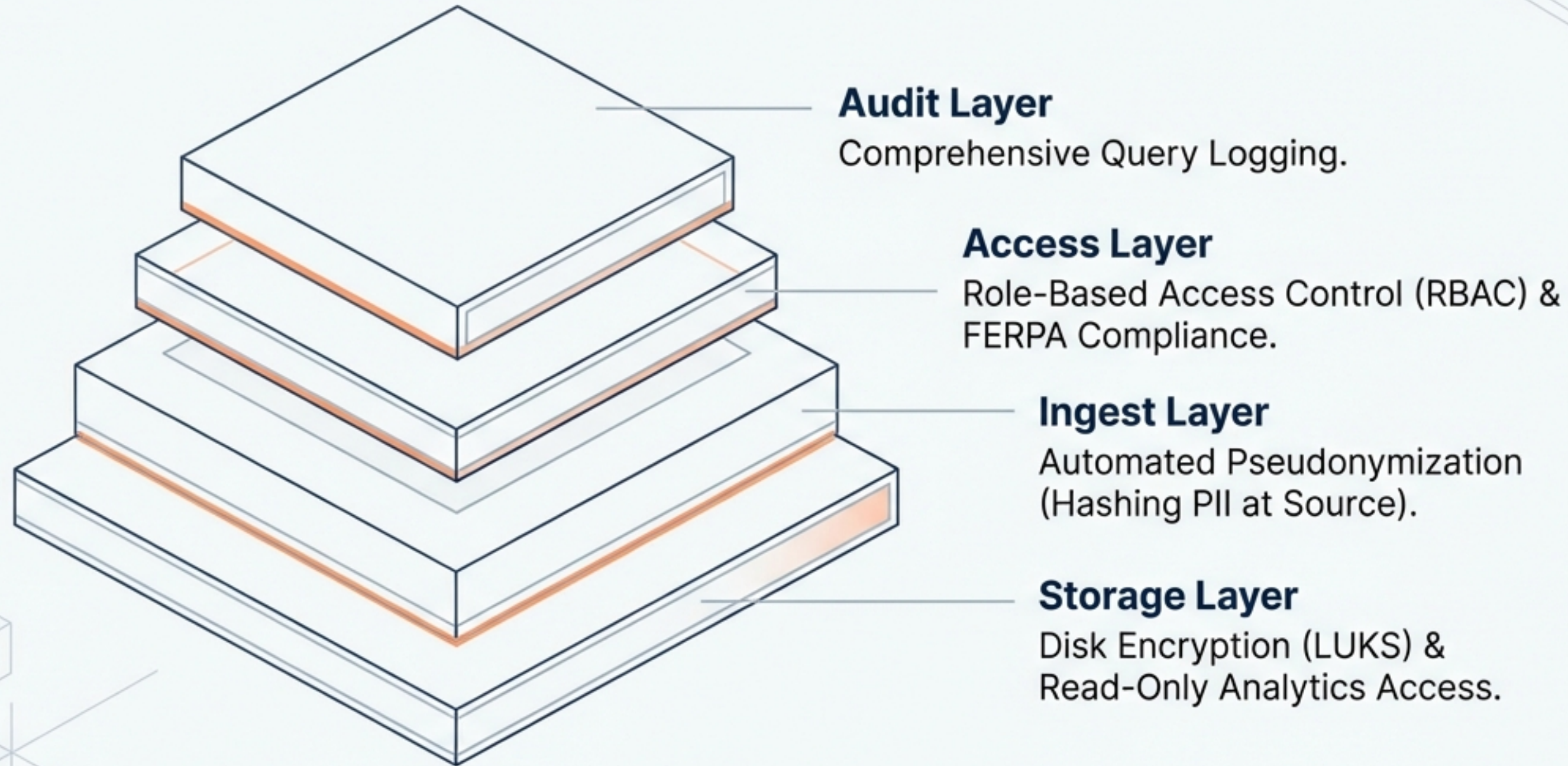


Connecting the silos allows us to see the ‘invisible’ students—those with passing grades but zero engagement, or high discipline risk but perfect attendance.

Equity Lens Callout:

Instantly spot achievement gaps across ELL, SPED, and FRL populations.

Security & Governance: Privacy by Design



Open source does not mean unsecured. Data stays strictly within district control. No third-party transfers.

The Implementation Pilot

Low Risk, High Reward. 90-Day Sprint.



Source Serif Pro:

- Setup Single Server (Docker)
- Connect SIS/LMS APIs

Source Serif Pro:

- Map Data to Ed-Fi Domains
- Validate Logic with dbt

Source Serif Pro:

- Launch Absenteeism Dashboard
- Train Counselors & Intervene



Technical Validity: Transparent & Reproducible



- [x] **Code-First:** Logic in version-controlled SQL/Python.
- [x] **Reproducible:** Audit logic from source to insight.
- [x] **Fairness:** Models tested for disparate impact.

Strategic Alignment: Built for Our Goals

District Goals:

Fiscal Responsibility

Digital Equity

Academic Excellence

OSS Outcomes:

→ \$158k/yr Savings & Zero Lock-in

→ Visibility into the Digital Divide

→ Real-time identification of At-Risk Students

This is not just an IT project.
It is a strategic enabler for the Superintendent's vision.

The New Standard.



Enterprise Intelligence.

Community Budget.

Student Success.

Source Serif Pro